

20-Week Project Roadmap

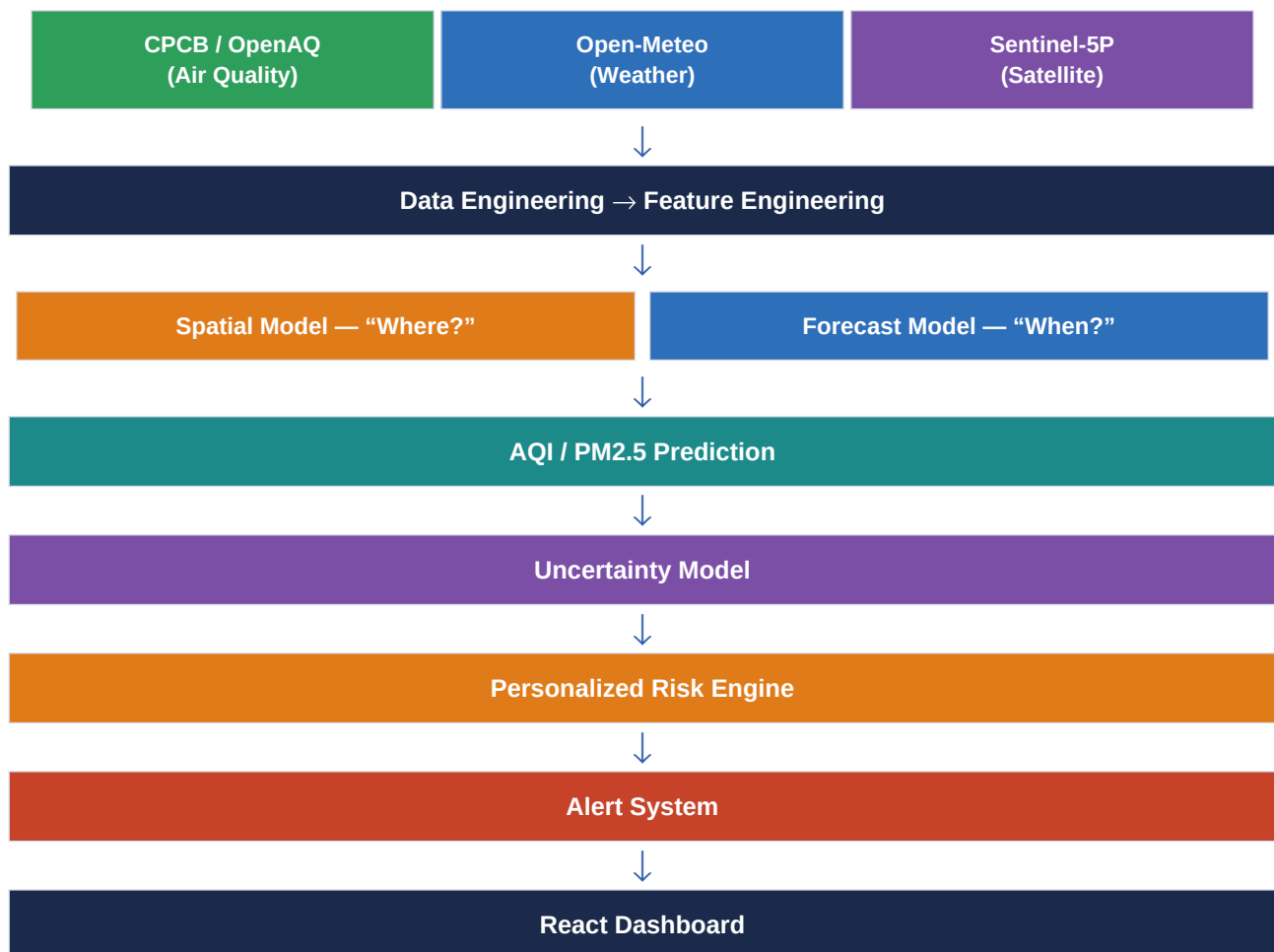
Hyperlocal Air Quality Forecasting + Personalized Health Risk Alerts

This document lays out a realistic, week-by-week plan for building the project over 4–5 months. Rather than combining every advanced idea (LSTM + GNN + satellite + live traffic) from day one, the project scope grows in stages — starting with a single city, a single pollutant, and short forecast horizons, then layering on spatial prediction, satellite data, uncertainty estimation, and personalization.

Core recommendation

Start with **one city/region, one main pollutant (PM2.5), and 1–6 hour forecasting**. Then add spatial prediction, satellite data, uncertainty, and personalization one step at a time. This keeps the project ambitious, but finishable.

Final System — High-Level Architecture



0. Recommended Scope Before You Start

Avoid: All of India + every pollutant + real-time traffic + satellite + GNN + TFT — that is too much for 20 weeks.

Instead, define a Phase 1 target:

Decision	Guidance
City	Pune or Ahmedabad recommended for manageable station coverage (Delhi / Mumbai also possible)
Main pollutant	PM2.5 first — then optionally PM10, NO ₂ , O ₃
Forecast horizon	Start at 1 hour ahead → extend to 3 hours → 6 hours
Spatial resolution	1 km × 1 km grid (or suitable resolution for the study area) — don't claim street-level accuracy

1. Data Source Stack

Data	Source	Purpose	Priority
PM2.5	CPCB / OGD	Main target	●●●●●
PM10	CPCB / OGD	Additional pollutant	●●● . .
NO ₂	CPCB / OGD	Feature	●●● . .
SO ₂	CPCB / OGD	Feature	●● . . .
CO	CPCB / OGD	Feature	●● . . .
O ₃	CPCB / OGD	Feature	●● . . .
Weather	Open-Meteo	Temp, humidity, wind, etc.	●●●●●
Satellite NO ₂	Sentinel-5P	Spatial feature	●●●● .
Satellite CO	Sentinel-5P	Spatial feature	●●● . .
Roads	OpenStreetMap	Spatial / traffic proxy	●●●● .
Population / land use	OSM / Census	Spatial features	●● . . .
Traffic	Optional API	Traffic features	●● . . .

1.1. Air-quality data — CPCB (primary ground truth)

The Government of India's Open Government Data platform provides real-time AQI data from monitoring stations, including PM2.5, PM10, NO₂, SO₂, CO and O₃, updated hourly. *Source: CPCB / India Open Government Data — Real-time AQI dataset.*

You need:

- station_id, station_name, latitude, longitude, timestamp
- PM2.5, PM10, NO₂, SO₂, CO, O₃

- AQI

Example row format:

Station	Lat	Lon	Time	PM2.5
Station A	18.52	73.85	10:00	75
Station A	18.52	73.85	11:00	82
Station B	18.55	73.90	10:00	110

1.2. OpenAQ — backup / secondary source

OpenAQ's API supports location-based and parameter/date-range queries, with both original measurements and hourly aggregations. Keeping OpenAQ as a backup means the project doesn't depend entirely on a single data source (CPCB = primary, OpenAQ = backup/validation).

1.3. Historical Kaggle dataset — initial development only

An 'India Air Quality Dataset 2015–2020' covers hourly/daily air-quality data across multiple Indian cities. Useful for Week 1–3 experimentation — but should not be the final data source; use official/primary data wherever possible for the final project.

1.4. Weather — Open-Meteo

Open-Meteo provides historical weather data and forecasts without requiring an API key for its non-commercial service, with hourly variables including temperature, humidity, precipitation, wind speed and wind direction.

You need:

- temperature, relative_humidity, wind_speed, wind_direction
- precipitation, surface_pressure, cloud_cover

1.5. Satellite — Sentinel-5P

Sentinel-5P/TROPOMI provides atmospheric observations including NO₂, CO, SO₂, O₃, HCHO and aerosol/cloud information via the Copernicus Data Space Ecosystem, which offers open, free Earth-observation data.

Important: Do not start with satellite data. First build the model with CPCB + Weather. Once that works, extend to CPCB + Weather + Satellite — this is much safer.

1.6. Traffic data

Don't make live Google traffic data a core dependency. Instead, start with traffic *proxy* features derived from time and geography, and only add live traffic later if time permits. Google's Routes API can provide route/route-matrix data, but it isn't necessary for the first ML model.

You need:

- hour_of_day, day_of_week, weekday/weekend, rush_hour
- road_density, distance_to_major_road, distance_to_highway, industrial_area

MVP: traffic proxy features → Advanced: real traffic API

1.7. Database

Development path: CSV → Parquet → PostgreSQL. Final recommendation: **PostgreSQL + PostGIS**, since the project is heavily geographic — storing station, latitude/longitude, pollution, weather, grid, road and

distance data together.

2. Week-by-Week Roadmap

Phase A — Weeks 1–2: Project Foundation

WEEK 1

Problem + Research

Learn about:

- AQI
- PM2.5, PM10, NO₂
- Spatial pollution
- Time-series forecasting
- Interpolation
- Uncertainty

Decide:

- City = _____
- Target = PM2.5
- Forecast = 1–6 hours
- Grid = _____

Deliverables:

- Problem statement
- Objectives
- Literature survey
- System architecture
- Dataset plan

What your guide should see: “We have finalized the problem, target city, target pollutant and data sources.”

WEEK 2

Dataset Collection

Set up a **data/** folder with raw/{cpcb, openaq, weather, satellite}, processed/, and final/ subfolders.

Download historical air-quality data — station latitude, longitude, timestamp, PM2.5, PM10, NO₂, etc.

Download weather data for the same stations/time range.

Phase B — Weeks 3–4: Data Engineering

WEEK 3

Cleaning

Expect: missing values, duplicates, wrong timestamps, outliers, mismatched units.

Handle:

- Missing values
- Duplicate rows
- Invalid readings
- Timestamp conversion
- Station consistency

Output: **clean_air_quality.csv**

WEEK 4

Data Integration

Join AIR + WEATHER data into a single Master Dataset, keyed by timestamp and station, with columns for lat, lon, PM2.5, PM10, NO2, temperature, humidity, wind_speed, wind_direction and rain.

Phase C — Weeks 5–6: Exploratory Data Analysis

WEEK 5

Pollution Analysis

Create graphs:

- PM2.5 vs time
- PM2.5 vs temperature
- PM2.5 vs humidity
- PM2.5 vs wind
- Hourly pollution pattern (e.g. a rush-hour spike around 7–9 AM)

WEEK 6

Spatial Analysis

Plot monitoring stations on a map and visualize which areas have higher pollution.

Also calculate:

- Distance between stations
- Distance to roads
- Number of nearby stations

Phase D — Weeks 7–8: Spatial Prediction (“Where?”)

WEEK 7

Baseline Spatial Models

Start with **Nearest Neighbor**, then implement **IDW (Inverse Distance Weighting)**: closer stations get more influence on the estimate for an unknown location (e.g. Station A = 80, Station B = 150 → weighted estimate in between, biased toward the closer station).

WEEK 8

Kriging

Implement **Ordinary Kriging** and compare Nearest Neighbor vs IDW vs Kriging using MAE and RMSE.

Deliverable: a hyperlocal pollution heatmap of the study area.

Phase E — Weeks 9–12: Forecasting

WEEK 9

Time-Series Feature Engineering

Lag features:

- $PM_{2.5}(t-1)$, $(t-2)$, $(t-3)$, $(t-6)$, $(t-12)$, $(t-24)$

Weather features:

- Temperature, humidity, wind, rain

Time features:

- Hour, day, month, weekday/weekend

Spatial features:

- Latitude, longitude, nearby_station_PM25

WEEK 10

Baseline ML Models

Don't jump straight to LSTM. Build a proper baseline first: Linear Regression, Random Forest, and XGBoost, then compare MAE / RMSE across the three.

WEEK 11

LSTM

Build sequence models: past 24 hours → LSTM → next 1 hour; then next 3 hours; then next 6 hours.

WEEK 12

Forecast Evaluation

Compare XGBoost vs LSTM using MAE, RMSE, R^2 , and MAPE where appropriate.

The report should answer: **which model predicts PM2.5 best?**

Phase F — Weeks 13–14: Satellite Integration

WEEK 13

Download / Process Satellite Data

Start with NO₂ only — don't try every Sentinel-5P variable at once. Extract satellite values around the study region into a table of latitude, longitude, date, NO₂.

WEEK 14

Combine Satellite + Ground Data

Train XGBoost on CPCB + Weather + Satellite and compare against the model without satellite data.

This is a strong experiment for the final report: if satellite data improves performance, that's evidence for it; if it doesn't, that's also a valid research finding.

Phase G — Weeks 15–16: Uncertainty

WEEK 15–16

Prediction Intervals

The model should output a prediction *and* an interval, e.g. “Predicted PM2.5 = 180, expected range 155–205, ~90% prediction interval.”

Approaches, easiest to hardest:

- Easy — Quantile Regression / gradient boosting quantile models
- Medium — Conformal Prediction (recommended)
- Advanced — Deep ensembles / MC Dropout

Recommended: Conformal Prediction — gives a strong uncertainty component without adding unnecessary complexity.

Phase H — Weeks 17–18: Personalization & Alerts

WEEK 17

Personalized Risk Engine

Build a user profile: age group, activity level, sensitivity category.

Avoid claiming the system diagnoses health conditions — it should provide informational pollution-risk alerts only (e.g. AQI = 90 → “Low/Moderate risk” for the general population, “Higher risk” for sensitive groups).

Risk engine inputs: AQI + Forecast + Uncertainty + User profile → Risk level.

WEEK 18

Alert System

Alert types:

- Current alert — “AQI is currently high”
- Forecast alert — “AQI expected to cross your alert threshold within 2 hours”
- Uncertainty alert — “Prediction confidence is low; expected range 170–230”

Start with dashboard notifications; add email or browser notifications if time allows. Don't spend significant time on SMS.

Phase I — Weeks 19–20: Dashboard, Testing & Presentation

WEEK 19

Full Dashboard

Page 1 — Home:

- Current AQI, PM2.5, risk level

Page 2 — Map:

- Interactive city map + AQI heatmap + monitoring stations

Page 3 — Forecast:

- Current, +1h, +2h, +3h, +4h, +5h, +6h

Page 4 — Personal Risk:

- User profile → risk level → recommendation

Page 5 — Model Info:

- Model used, MAE, RMSE, prediction interval, confidence

WEEK 20

Testing + Presentation

Test data handling:

- Missing values
- API failure
- Bad data

Test ML:

- Overfitting
- Data leakage
- Forecast accuracy

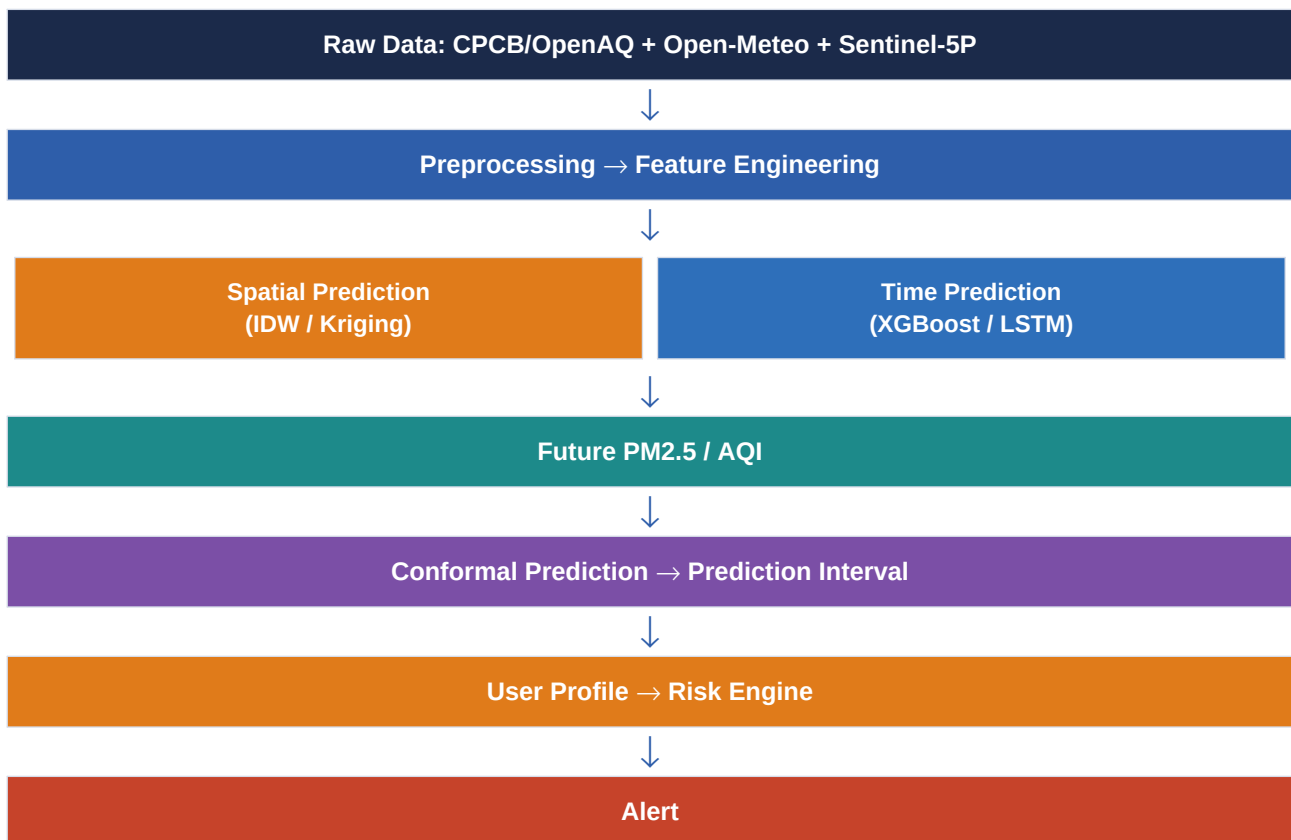
Test application:

- Wrong location
- No sensor nearby
- API unavailable

Final presentation flow:

- Problem → Existing limitations → Our solution → Architecture → Datasets → ML models → Results → Dashboard → Limitations → Future work

3. Final ML Pipeline



4. Recommended Repository Structure

```
hyperlocal-air-quality/
├── data/
│   ├── raw/ (cpcb, openaq, weather, sentinel5p)
│   ├── processed/
│   ├── features/
│   └── notebooks/
│       ├── 01_data_collection.ipynb
│       ├── 02_data_cleaning.ipynb
│       ├── 03_eda.ipynb
│       ├── 04_spatial_analysis.ipynb
│       ├── 05_idw.ipynb
│       ├── 06_kriging.ipynb
│       ├── 07_xgboost.ipynb
│       ├── 08_lstm.ipynb
│       ├── 09_satellite.ipynb
│       └── 10_uncertainty.ipynb
├── src/
│   ├── data/
│   ├── preprocessing/
│   ├── spatial/
│   ├── forecasting/
│   ├── uncertainty/
│   ├── personalization/
│   ├── models/ (xgboost, lstm, spatial)
│   ├── backend/ (FastAPI)
│   ├── frontend/ (React)
│   ├── tests/
│   ├── requirements.txt
│   ├── README.md
│   └── docker-compose.yml
```

5. API / Data Source Summary

Must-have

Source	Purpose
CPCB / India OGD	Ground air-quality data (real-time AQI dataset).
OpenAQ	Air-quality measurements — validation / backup.
Open-Meteo	Weather data; no API key required for non-commercial use.
Sentinel-5P	Satellite atmospheric pollution data (via Copernicus).
OpenStreetMap	Roads, road density, industrial areas, distance to highways, geographic features.

Optional

Google Maps Routes API — use only if you want additional traffic/route-related features; not required for the first ML model.

6. Final Master Dataset Schema

This is the most important dataset in the project. After joining all sources, rows should look approximately like this:

time	lat	lon	PM2.5	PM10	NO2	temp	hum	wind	rain	sat_N O2	road_d ens	hr	target_ 1h
10:00	18.52	73.85	85	120	42	30	65	3.2	0	0.0001	4.5	10	92
11:00	18.52	73.85	92	125	45	31	63	2.8	0	0.0002	4.5	11	105

The final target column is **target_PM25_1h** — “what will PM2.5 be one hour from now?” Later, add **target_PM25_3h** and **target_PM25_6h** for longer horizons.

A note on AQI vs. PM2.5

The model should predict **PM2.5 concentration** first, rather than directly predicting AQI, since PM2.5 is a physical pollutant concentration while AQI is an index calculated from it. Pipeline: Model predicts PM2.5 → PM2.5 concentration → AQI calculation → AQI category. This makes the methodology much easier to explain.

7. Suggested Experiments

Running a structured set of experiments — rather than a single model — is what makes this look like a real research project rather than just an application.

Model	Weather	Spatial	Satellite	MAE	RMSE
Baseline	No	No	No	—	—
XGBoost	Yes	No	No	—	—
XGBoost	Yes	Yes	No	—	—
XGBoost	Yes	Yes	Yes	—	—
LSTM	Yes	No	No	—	—

Do not invent these numbers in advance — fill the MAE/RMSE columns in after actually running each experiment.

8. What Is Actually Mandatory?

With only 20 weeks, prioritize ruthlessly. A completed core system is far better than an incomplete ambitious one.

MANDATORY

CPCB/OpenAQ + Weather + Data cleaning + EDA + XGBoost baseline + LSTM + Spatial interpolation + Dashboard.

RECOMMENDED

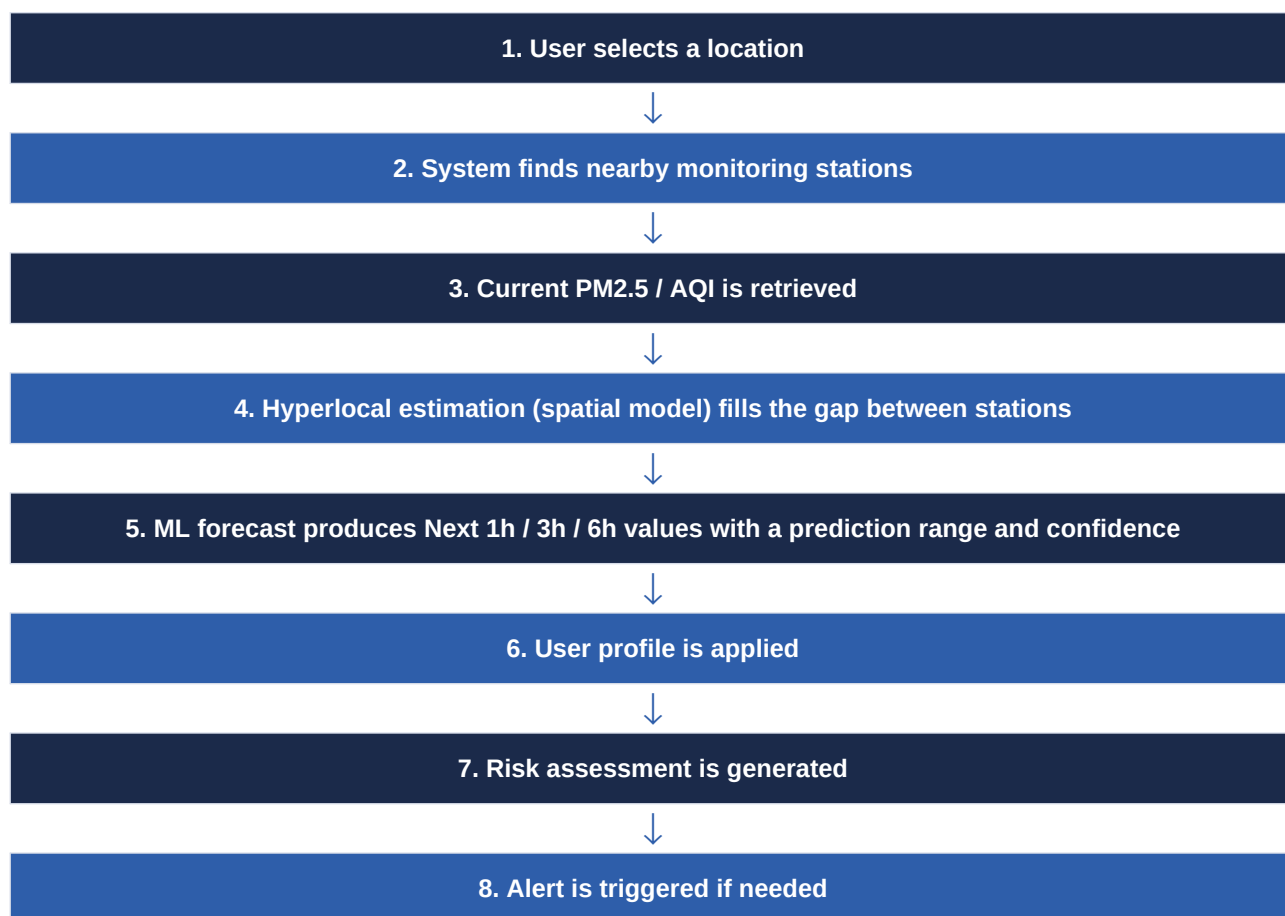
Sentinel-5P satellite data + Uncertainty (conformal prediction) + Personalization.

IF TIME REMAINS

GNN, TFT, live traffic, real-time streaming, mobile app, SMS alerts.

A completed **CPCB + Weather + Spatial Model + XGBoost/LSTM + Uncertainty + Personalized Dashboard** is much better than an incomplete **CPCB + Weather + Satellite + GNN + TFT + Kafka + React + Mobile App**.

9. Final Demonstration Flow



Core research question

“Can we improve neighborhood-level short-term PM2.5 forecasting by combining sparse ground observations with weather, spatial information and satellite observations, while also communicating prediction uncertainty and personalized risk?”

This is a substantially stronger framing than simply “Can we predict AQI using LSTM?”

10. Recommended Final Stack

Layer	Choice
Data	CPCB/India OGD + OpenAQ + Open-Meteo + Sentinel-5P + OpenStreetMap
ML	XGBoost → LSTM
Spatial	IDW → Kriging
Uncertainty	Conformal Prediction
Backend	FastAPI
Database	PostgreSQL + PostGIS

Layer	Choice
Frontend	React + Leaflet / Mapbox
Deployment	Docker
Primary target	PM2.5
Forecast horizon	1–6 hours
Study area	One Indian city, initially

This scope is ambitious enough to impress — while still being realistically finishable within a 20-week timeline.